

Ques 1 : Arithmetic Precedence

Goal: Understand the order of operations in complex arithmetic expressions.

Instructions:

1. Declare a variable x using let and set it to 10.
2. Declare a variable y using let and set it to 5.
3. Declare a variable result and assign it the value of the following expression: $x + y * 2 - 4 / 2$.
4. Print the final result.

Expected output : 18

Ques 2 : Strict vs. Loose Equality

Goal: Highlight the difference between the loose (==) and strict (===) equality operators.

Instructions:

1. Declare a const variable numStr with the value "42".
 2. Declare a const variable numVal with the value 42.
 3. Print the result of `numStr == numVal`.
 4. Print the result of `numStr === numVal`.
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Ques 3 : Logical AND Operator (&&)

Goal: Use the logical AND operator to check multiple conditions and practice short-circuiting.

Instructions:

1. Declare a let variable age and set it to 25.
 2. Declare a let variable hasID and set it to true.
 3. Declare a variable canEnter and set it to the result of checking if age is greater than 21 **AND** hasID is true.
 4. Print the boolean value of canEnter.
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Ques 4 : Modulo Operator (%)

Goal: Use the modulo operator to determine if a number is even or odd.

Instructions:

1. Declare a const variable numberToCheck and set it to 13.
2. Declare a variable isEven and set it to the result of the comparison: `numberToCheck modulo 2 equals 0`.
3. Print the boolean value of isEven.

Ques 5 : Compound Assignment Operator (+=)

Goal: Practice using the compound addition assignment operator.

Instructions:

1. Declare a let variable total and set its initial value to 100.
 2. Use the += operator to add 50 to total.
 3. Use the += operator to add the string " USD" to the updated total. (Observe the type coercion).
 4. Print the final total.
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Ques 6 : The typeof Operator

Goal: Use the typeof operator to check the datatype of different values.

Instructions:

1. Declare a const variable dataA with the value 99.9.
 2. Declare a const variable dataB with the value null.
 3. Print the result of typeof dataA.
 4. Print the result of typeof dataB.
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Ques 7 : Logical NOT Operator (!)

Goal: Practice using the logical NOT operator to flip a boolean value and coerce a number to a boolean.

Instructions:

1. Declare a let variable isComplete and set it to false.
 2. Print the result of !isComplete.
 3. Print the result of !0 (Coercing the number 0).
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Ques 8 : Increment and Decrement Operators

Goal: Understand the post-fix increment operator (++).

Instructions:

1. Declare a let variable count and set it to 5.
2. Declare a let variable newCount and assign it the value of count++.
3. Print the value of newCount.
4. Print the final value of count.

Ques 9 : Prefix Increment Operator (++)

Goal: Understand the prefix increment operator, where the increment happens *before* the value is assigned.

Instructions:

1. Declare a let variable points and set it to 10.
 2. Declare a let variable bonusPoints and assign it the value of ++points.
 3. Print the value of bonusPoints.
 4. Print the final value of points.
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Ques 10 : Logical OR Operator (||) and Defaults

Goal: Use the logical OR operator to assign a default value.

Instructions:

1. Declare a const variable username and assign it the value "" (an empty string).
 2. Declare a const variable defaultName with the value "Guest".
 3. Declare a variable displayName and assign it the first truthy value between username OR defaultName.
 4. Print the displayName.
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Ques 11 : Exponentiation Operator ()**

Goal: Use the exponentiation operator for calculating powers.

Instructions:

1. Declare a const variable side and set it to 3.
2. Calculate the volume of a cube (side cubed) using the ** operator and store it in a variable named volume.
3. Print the volume with the message: "Cube volume is [value]".

Sample input : side = 3

Expected output : cube volume is 27

Ques 12 : Type-Based Conditional Output

Goal: Combine typeof with the comparison operators.

Instructions:

1. Declare a const variable input and set its value to true.
 2. Declare a variable isBoolean and assign it the result of checking if the typeof input is strictly equal (===) to the string "boolean".
 3. Print the boolean value of isBoolean.
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Ques 13 : Division and Remainder in One Expression

Goal: Perform sequential arithmetic operations to calculate two different results from a single input.

Instructions:

1. Declare a const variable totalSeconds and set it to 130.
 2. Declare a variable minutes and assign it the whole number result of dividing totalSeconds by 60 (you can use Math.floor() or simply division).
 3. Declare a variable remainingSeconds and assign it the remainder of totalSeconds divided by 60 using the modulo operator (%).
 4. Use a template literal to print the result: "130 seconds is 2 minutes and 10 seconds." (Adjust the numbers based on your calculated variables).
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Ques 14 : Type Coercion with + and Unary Plus

Goal: Use the unary plus operator (+) to explicitly coerce a string to a number.

Instructions:

1. Declare a const variable stringA with the value "10".
 2. Declare a const variable stringB with the value "5".
 3. Declare a variable numericSum and calculate the arithmetic sum of the two variables by prefixing each with the unary plus operator (+stringA + +stringB).
 4. Print the numericSum.
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Ques 15 : Checking for Integers

Goal: Use the built-in Number.isInteger() method to check the datatype property.

Instructions:

1. Declare a const variable val1 with the value 10.0.
 2. Declare a const variable val2 with the value 10.5.
 3. Print the boolean result of checking if val1 is an integer using Number.isInteger().
 4. Print the boolean result of checking if val2 is an integer using Number.isInteger().
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